Complex Analysis Homework 8

1. Find the Laurent series of $\frac{1}{z+1}+\frac{1}{z+i}$ at 1 from $(z-1)^{-2}$ to $(z-1)^{2}$ in each of the following domains.

(a) the region $0<|z-1|<\sqrt{2}$;

(b) the region $\sqrt{2}<|z-1|<2$;

(c) the region $|z-1|>2$.

2. Evaluate the following integrals with counterclockwise contour using residue theorem.

(a) $\int_{|z|=1}\frac{\sin z}{z^{2}}dz$

(b) $\int_{|z|=2}\frac{e^{z}}{z^{2}-1}dz$

(c) $\int_{|z|=2}\frac{z}{\cos z}dz$

(d) $\int_{|z|=1}\frac{z^{4}}{\sin z}dz$

(e) $\int_{|z-1|=1}\frac{1}{z^{8}-1}dz$

(f) $\int_{|z-1/2|=3/2}\frac{\tan z}{z}dz$

3. Using residue theory, calculate the following integrals:

(a) $\int_0^{2\pi} \frac{d\theta}{a+b\sin\theta}, \quad a > b > 0$

(b) $\int_0^{2\pi} \frac{d\theta}{1+3\cos^2\theta}$

(c) $\int_0^{\pi} \frac{\cos 2\theta}{1-2a\cos\theta+a^2} d\theta$ where $a^2 < 1$.

(d) $\int_0^{2\pi} \frac{\cos 2\theta}{5-4\cos\theta} d\theta$

(e) $\int_{-\infty}^{\infty} \frac{dx}{(x^2+a^2)^2}$